

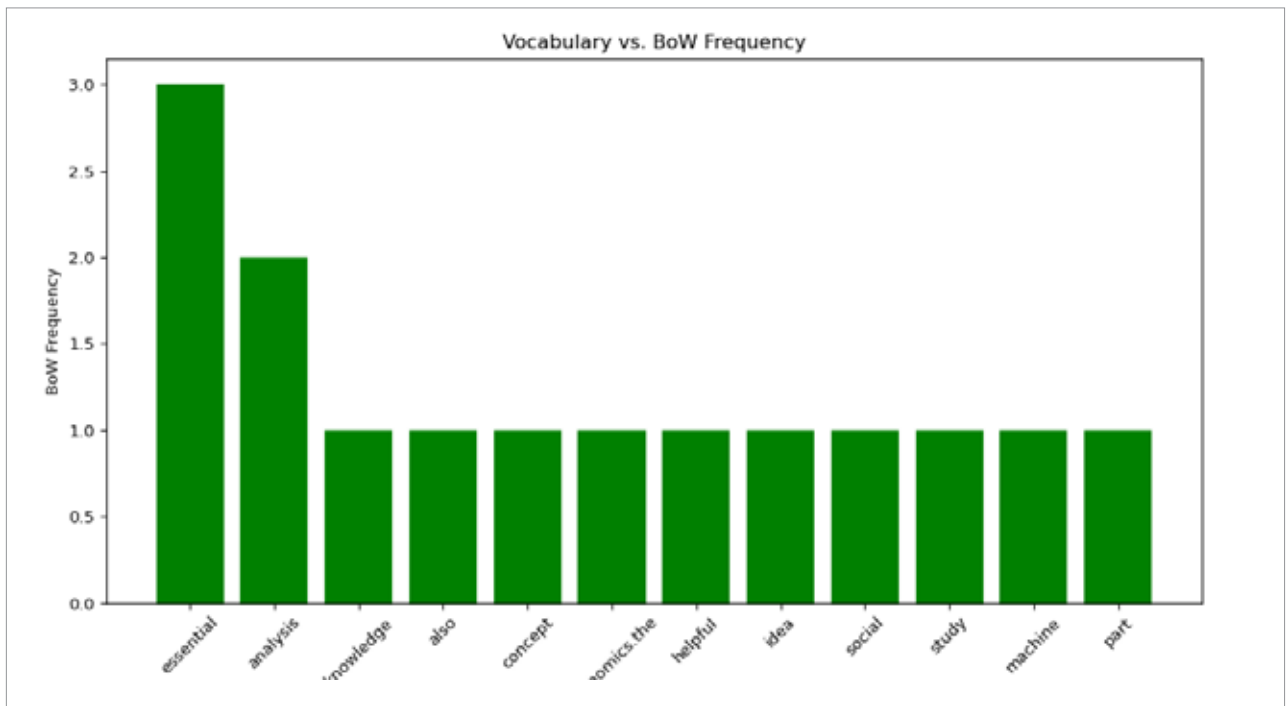


Figure 2: Word frequency plot of the corpus

```
vocabulary2 = [dictionary[id] for id in dictionary.keys()]
# Flatten the corpus to count word frequencies
word_counts = Counter()
for doc in corpus:
    for word_id, freq in doc:
        word = dictionary[word_id]
        word_counts[word] += freq
        print(word_id, freq, word, word_counts[word])

## 1 1 analysis 1
## 6 1 essential 1
## 9 1 knowledge 1
## 0 1 also 1
## 2 1 concept 1
## 5 1 economics.the 1
## 6 1 essential 2
## 7 1 helpful 1
## 8 1 idea 1
## 13 1 social 1
## 15 1 study 1
## 1 1 analysis 2
## 6 1 essential 3
## 11 1 machine 1
## 12 1 part 1
# Convert to a DataFrame for easier visualization
df = pd.DataFrame(word_counts.items(), columns=['Word',
'Frequency'])
df = df.sort_values(by='Frequency', ascending=False)
# Plotting the frequency graph
```

```
plt.figure(figsize=(12, 6))
plt.bar(df['Word'], df['Frequency'], color = 'green') # Show
top 20 words
plt.title('Vocabulary vs. BoW Frequency', fontsize=12)
plt.xlabel('Vocabulary')
plt.ylabel('BoW Frequency')
# set font size for both major and minor ticks
plt.xticks(rotation=45, fontsize = 10)
## ([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11], [Text(0,
0, 'essential'), Text(1, 0, 'analysis'), Text(2, 0,
'knowledge'), Text(3, 0, 'also'), Text(4, 0, 'concept'),
Text(5, 0, 'economics.the'), Text(6, 0, 'helpful'), Text(7,
0, 'idea'), Text(8, 0, 'social'), Text(9, 0, 'study'),
Text(10, 0, 'machine'), Text(11, 0, 'part')])
plt.show()
```

As can be seen, the corpus's vocabulary and word frequency are provided by the BoW model in conjunction with corpora. Dictionary. Additionally, it associates words with their integer identification numbers, which serves as the foundation for text processing and advanced corpus comprehension. END 🐧

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The author has more than 30 years of experience in software development. He has published two books on MATLAB and medical image processing (published by PHI Learning and SPD).